Unit 9, Lesson 6: Describing and Interpreting Histograms – Part 2

Benchmark

MA.6.DP.1.4 Given a histogram or line plot within a real-world context, qualitatively describe and interpret the spread and distribution of the data, including any symmetry, skewness, gaps, clusters, outliers, and the range.

Learning Targets

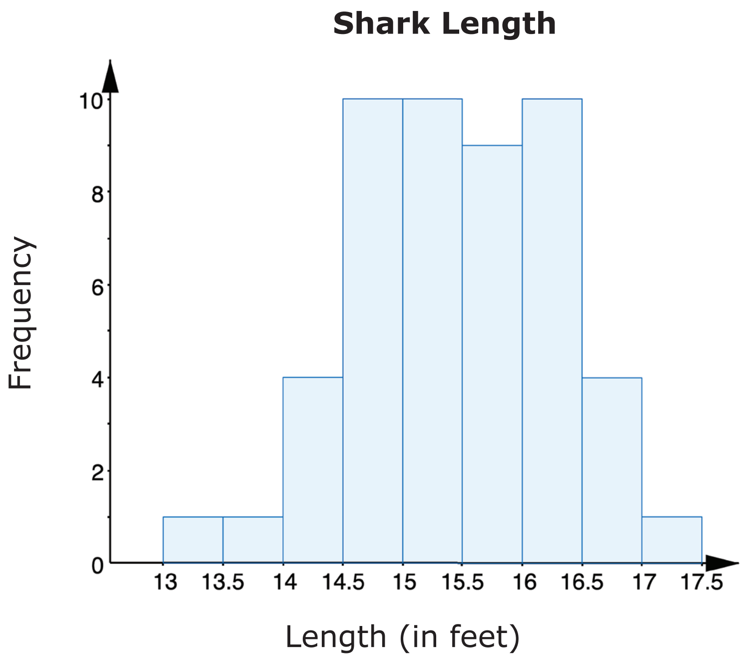
- ☐ I can describe the shape of a histogram.
- ☐ I can describe the spread of a data set given a histogram.
- ☐ I can describe gaps, outliers, and clusters in a histogram.

9.6.1 Warm-Up: Math Fail

1. Discuss with your partner what you notice about this picture.

9.6.2 Guided Instruction: Shape of Distributions

1. Consider the histogram of the lengths of female sharks.



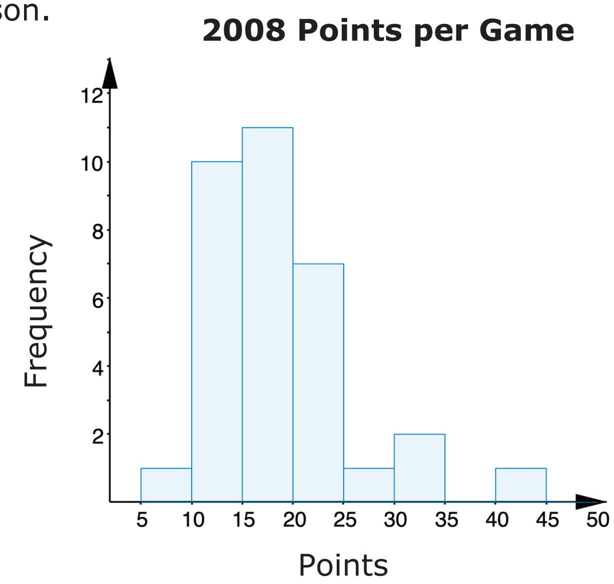
a. Sketch an approximate line of symmetry for the histogram.

b. If the histogram is folded along the line of symmetry, will each half line up perfectly?

c. Do you think all histograms will be symmetrical?

The shape of the shark lengths histogram can be described as **symmetric**, but histograms can also have other shapes.

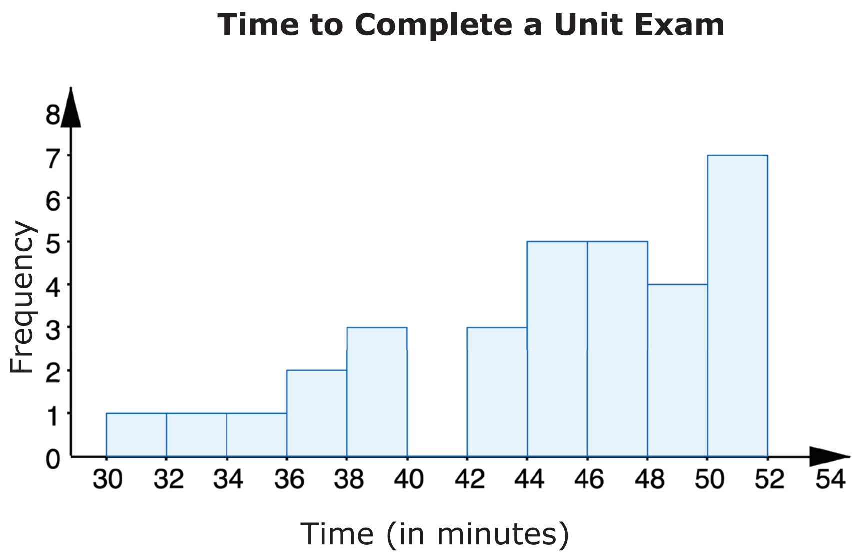
2. The histogram displays the number of points per game scored by a college basketball player during the 2008 season.



Discuss with your partner what you notice about the shape of the distribution of points scored per game.

A ***right-skewed*** graph has a longer trail of data to the right and the peak of the data on the left.

3. The time to complete a unit exam for 32 students is shown on the histogram.

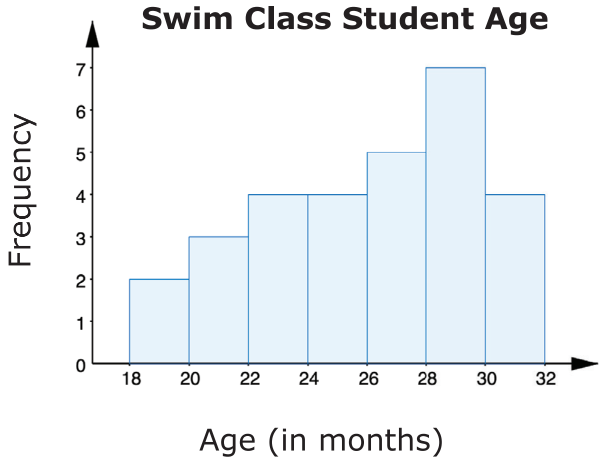


Write an observation about the shape of the distribution of time spent to complete the exam.

A ***left-skewed*** graph has a longer trail of data to the left and the peak of the data on the right.

9.6.3 Guided Practice: Describing Shape of Distributions

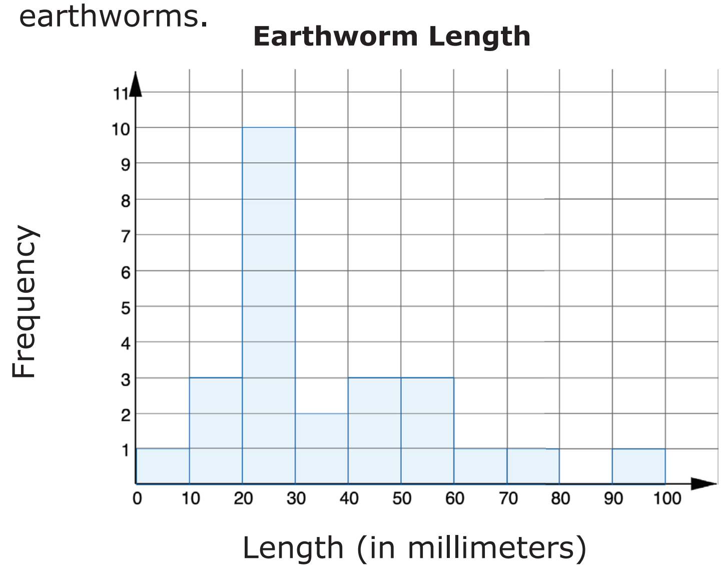
1. This histogram shows the ages of students in a swimming class.



The shape of the distribution of ages can be best described as

- ○ symmetric.
- ○ right skewed.
- ○ left skewed.

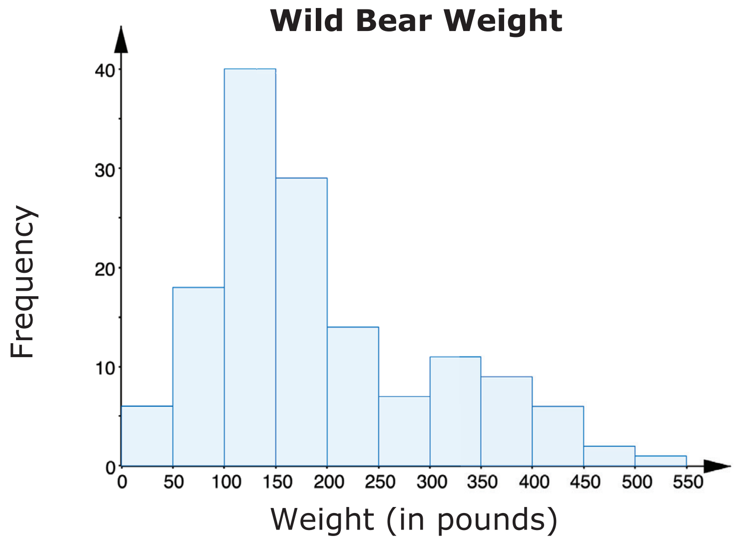
2. The histogram shows the length, in millimeters, of 25 earthworms.



The shape of the distribution of the earthworms' lengths can be best described as

- ○ symmetric.
- ○ right skewed.
- ○ left skewed.

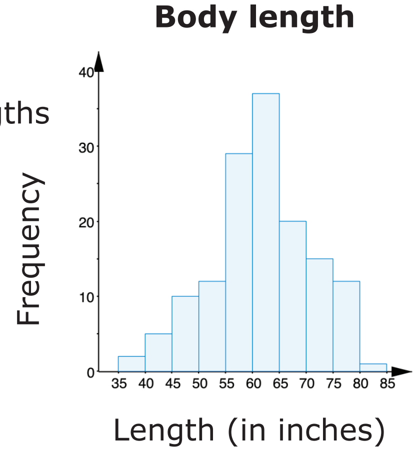
3. In one study of wild bears, researchers measured the weights, in pounds, of 143 wild bears that ranged in age from newborn to 15 years old. The data were used to make this histogram.



Describe the shape of the graph given the context.

4. The histogram summarizes the data on the body lengths of the same 143 wild bears.

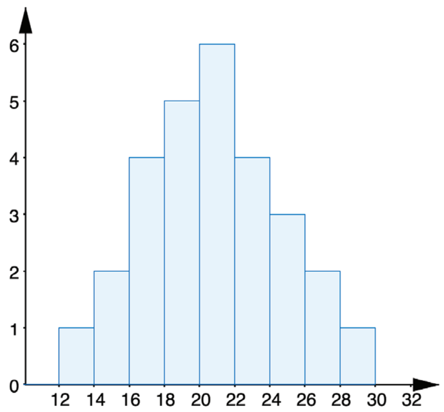
Describe the shape of the distribution of the lengths of bears.



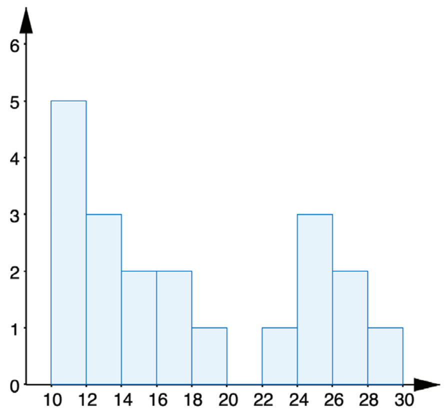
9.6.4 Guided Instruction: Features of Histograms

Below are two histograms with different shapes and features.

Histogram A



Histogram B



1. Histogram A is
   - ○ symmetrical
   - ○ not symmetrical

   and has a peak near ______.

2. Histogram B is
   - ○ symmetrical
   - ○ not symmetrical

   and has
   - ○ one peak.
   - ○ more than one peak.

A ***cluster*** is formed when data points are in a close group or numerical data have similar values on a number line.

3.
   - ○ Histogram A
   - ○ Histogram B

   has two clusters.

A ***gap*** shows a location with no data values. On a histogram, a space between the bins indicates a gap, or a frequency of zero for that bin.

4.
   - ○ Histogram A
   - ○ Histogram B

   has a gap between _____ and _____.

An ***outlier*** is a value that is much higher or much lower than the other values in a data set.

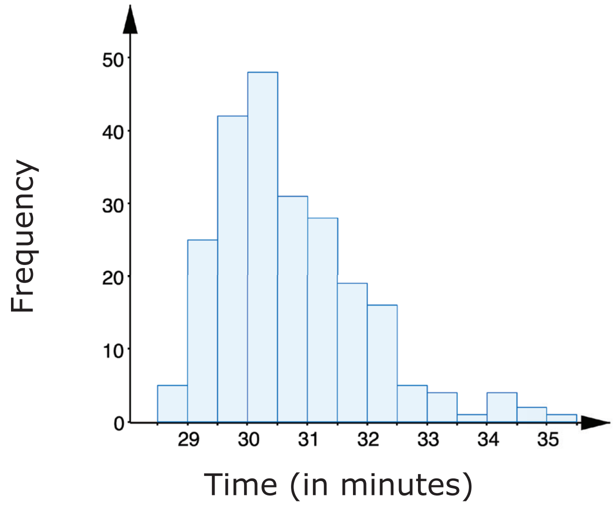
5. If the data value
   - ○ 12
   - ○ 24
   - ○ 40

   were added to Histogram A, it would be considered an outlier.

9.6.5 Collaboration: Features of Histograms

1. Daveed has been tracking the time, in minutes, it takes him to run 4 miles for the last 10 years. A histogram of Daveed's run times is shown.

Daveed's Run Times

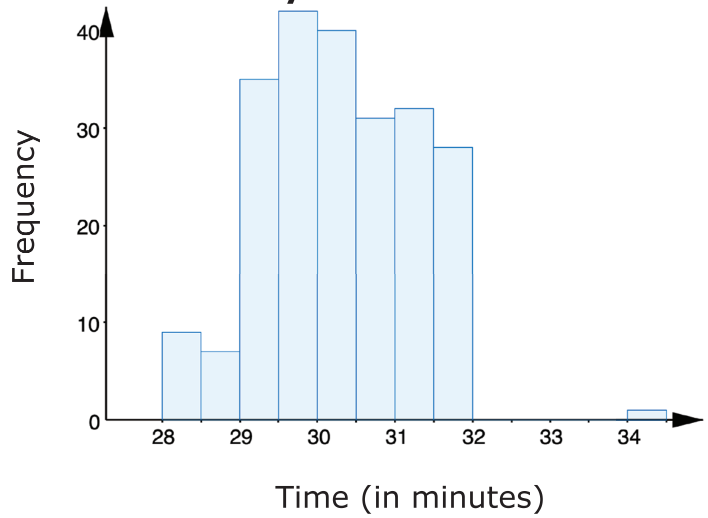


a. The shape of run times can be best described as
   - ○ symmetrical.
   - ○ right skewed.
   - ○ left skewed.

b. Daveed's typical run time was about __________ minutes. Most of his run times were between _______ to _______ minutes. Not many of his runs were ○ shorter ○ longer than 32.5 minutes.

2. Arya used the data of her 4-mile run times to create a different histogram.

Arya's Run Times

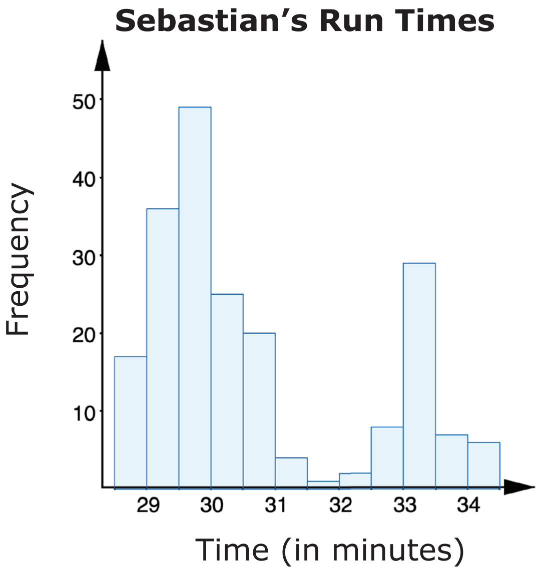


a. Discuss with your partner what you notice about the interval widths. Determine if there are any unusual features including clusters, gaps, or outliers. Write a detailed description of your discussion.

b. At what interval in the histogram of Arya's times is there an outlier?

c. Would this same data value also be considered an outlier in the histogram of Daveed's times?

3. Sebastian used the data of his 4-mile run times to create a third histogram.



a. Which feature is depicted in Sebastian's histogram?

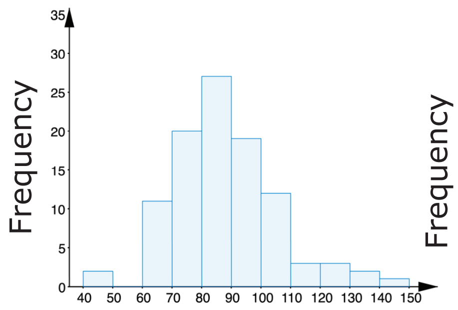
b. Describe the spread and shape of the distribution of Sebastian's run times.

4. Different data will change the intervals for the horizontal axis and will affect the histogram and its features: symmetry, skewness, gaps, clusters, outliers, and the range. With your partner, review all three run-time scenarios and discuss the differences between the histograms.

9.6.6 Your Turn!

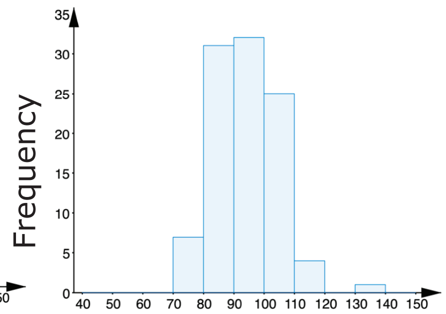
1. The histograms show the number of text messages sent in one week by a group of 100 sixth-grade and 100 seventh-grade students.

Sixth Graders



Number of Text Messages

Seventh Graders



Number of Text Messages

a. Describe the shape of the histogram for the number of text messages sent in one week by sixth-grade students.

b. Which group of students has the larger spread?

c. Explain if sixth graders or seventh graders tend to be more alike in terms of the number of text messages they send.

9.6.7 Wrap-Up: Reflection

Place an **X** on the line to indicate your understanding of parts of analyzing a histogram. Then provide a brief explanation or evidence to support your self-assessment.

1. I can describe the shape of a histogram.

I need help. I can't get started.	I'm getting there.	I understand and have evidence.

2. I can describe gaps, outliers, and clusters in a histogram.

I need help. I can't get started.	I'm getting there.	I understand and have evidence.

3. I can describe the spread of a data set given a histogram.

I need help. I can't get started.	I'm getting there.	I understand and have evidence.